ClinicOS

Local-first consultation documentation. ClinicOS records the consultation that is already happening, transcribes it and drafts the clinical note on the clinic's own computer, and the doctor only reviews and signs. This sheet covers the workflow, the architecture and every dependency.

Status	Working prototype with v1 accounts, roles and dashboards. Verified by `npm run test:e2e` (153 checks over HTTP).
Stack	Next.js 15 · React 19 · TypeScript · SQLite via better-sqlite3 · faster-whisper · Ollama
Full reference	13-page version: `artifacts/ClinicOS_Prototype_Documentation.pdf`

01 What it does

A doctor in a busy clinic gets roughly two minutes per patient, and the record is what gets dropped. ClinicOS captures the consultation, transcribes it locally, drafts a structured note, and asks the doctor only to review and sign. A branded A4 prescription prints from the signed note. Every stage runs on the clinic machine, with no cloud service and no internet needed at run time.

> **Design boundary.** ClinicOS stores, formats and communicates clinical data. It generates no diagnoses, risk scores or triage flags, which keeps it outside medical-device regulation under India's CDSCO rules. The drafting model's system prompt enforces this.

02 Consultation workflow

Four screens (Consent → Record → Transcript → Review & sign) over a six-stage data lifecycle.

STAGE 1
Consent
The doctor reads the consent script. Recording endpoints return 403 until `consentGiven` is set: a server-side gate, not a UI one.

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STAGE 2
Record
The browser streams audio to disk in 5-second WebM chunks, so a crash or power cut loses at most five seconds.

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STAGE 3
Transcribe
Chunks go to the local faster-whisper sidecar. Language is auto-detected, which handles Bengali–English code-mixing.

STAGE 4
Draft
Ollama (qwen2.5:7b-instruct) turns the transcript into `ClinicalNote` JSON, stored as `draftJson` and never overwritten.

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STAGE 5
Review & sign
Edits go to `workingJson`, autosaved to the server and localStorage. Signing writes `finalJson`, deletes the audio and logs each changed field.

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STAGE 6
Print
An A4 prescription renders from the signed note, carrying the clinic and doctor details from Clinic admin.

Accuracy metric	The diff between `draftJson` and what the doctor signs measures how good the drafting really is, with no self-reporting.
Recovery	Unfinished consultations reappear on Today with the reason they stopped. A truncated WebM still decodes up to the cut, so partial audio is never lost.

03 Runtime architecture

Three processes on the clinic machine. None accepts a connection from outside it.

PORT	PROCESS	ROLE
`:3000`	**Next.js app** (App Router)	Interface and all 27 API routes. Owns the SQLite database through better-sqlite3.
`:8756`	**ASR sidecar** (FastAPI + uvicorn)	faster-whisper, `small` model, int8 on CPU. A voice-activity filter drops waiting-room silence.
`:11434`	**Ollama** (native app)	qwen2.5:7b-instruct drafts the structured note.

Providers	`AsrProvider` (local | http) and `LlmProvider` (ollama | openai). An OpenAI-compatible hosted endpoint can replace either through `.env` alone. Components and routes never call Whisper or Ollama directly.
Storage	SQLite with hand-written SQL and no ORM. Schema, idempotent migrations (`PRAGMA table_info` then a conditional `ALTER TABLE`) and every query live in `src/lib/db.ts`.
Backups	SQLite's online backup API, never a file copy: in WAL mode a copy silently misses recent writes. A daily snapshot is taken on page load, 14 are kept, and restore sits behind a typed `RESTORE`.

04 Data model

Eight tables. Consultation content is sensitive personal data under India's DPDP Act, 2023; the database, its WAL files, backups and audio are all kept out of version control.

TABLE	HOLDS
`Patient`	Name (indexed for search), age, sex, phone, guardian.
`Consult`	Consent (who, when), audio path and metrics, `audioDeletedAt`, transcript, `draftJson` (immutable), `workingJson`, `finalJson`, model names and timings, `signedAt`.
`Edit`	Every field the doctor changed from the draft, before and after.
`User`	Accounts: role (doctor, staff, admin), `isAdmin`, scrypt password hash.

TABLE	HOLDS
`Session`	SHA-256 hashes of session tokens only; a 12-hour sliding idle limit.
`AuditEvent`	The activity log. Never holds a patient's name.
`Setting`	Clinic identity edited in Clinic admin. `.env` is only the fallback.
`QueueEntry`	The front desk's numbered list for the clinic computer's local day.

ClinicalNote — `chiefComplaint`, `history`, `examination`, `diagnosis`, `medications[]` (name, dose, frequency, duration, instructions), `investigations[]`, `advice`, `followUp`. `parseNote()` is defensive: malformed model output degrades to an empty note instead of throwing.

05 Accounts and permissions

CAPABILITY	DOCTOR	FRONT DESK	CLINIC ADMIN
clinical — transcripts, notes, signing, prescriptions, patient history	Yes	—	—
frontdesk — find and register patients, today's list, record consent	Yes	Yes	—
admin — people, clinic details, backups, export, deletion, activity log	If `isAdmin`	—	Yes

Every page calls `requireUser()` and every route `requireApiUser(capability)`: hiding a link is never the control. `/setup` creates the owner on first run, then refuses forever.

06 API surface

27 route handlers under `src/app/api`, all local-only.

GROUP	ROUTES
Consultation (9)	`/api/consults` · `[id]/consent` · `[id]/audio` · `[id]/audio/chunk` · `[id]/audio/finalize` · `[id]/transcribe` · `[id]/draft` · `[id]/note` · `[id]/sign`
Patients (3)	`/api/patients` · `[id]` · `[id]/export`
Today's list (3)	`/api/queue` · `[id]/start` · `[id]/left`
Backup (4)	`/api/backup/create` · `download` · `restore` · `export`
Accounts (8)	`/api/setup` · `/api/auth/login` · `logout` · `password` · `/api/admin/settings` · `users` · `users/[id]` · `users/[id]/reset`

07 Dependencies

Node runtime — `clinicos-app/package.json`. One Next.js project ships the interface and the API together.

PACKAGE	VERSION	TIER	PURPOSE
`next`	^15.1.0	BOTH	App Router framework: the UI and every API route handler.
`react · react-dom`	^19.0.0	FE	UI runtime and DOM renderer.
`better-sqlite3`	^13.0.3	BE	Synchronous SQLite driver with the online backup API. **Keep on v13+**: its N-API prebuilds need no compiler on any Node version.
`motion`	^13.2.0	FE	Functional UI motion, under 200 ms on clinical screens.
`gsap · lenis`	^3.15.0 · ^1.3.26	FE	Timelines and smooth scroll. Marketing surfaces only.
`three · @react-three/fiber · drei`	^0.186 · ^9.7 · ^10.7	FE	3D, loaded through `next/dynamic` with `ssr: false`. Never on clinical screens.
`typescript · @types/*`	^5.7.0	BUILD	Type checking: node, react, react-dom, better-sqlite3, three.

Python sidecar — `asr/requirements.txt`

PACKAGE	VERSION	PURPOSE
`faster-whisper`	≥1.1.0	CTranslate2 Whisper inference: the transcription itself.
`fastapi · uvicorn`	≥0.115 · ≥0.32	The sidecar's HTTP interface and ASGI server.
`python-multipart`	≥0.0.12	Multipart parsing for audio uploads.

External prerequisites — installed once, located by the launchers

Node.js LTS · Python 3	Windows: `winget install OpenJS.NodeJS.LTS` and `winget install Python.Python.3.12`. The launcher creates the sidecar's virtualenv.
Ollama	Native app, then `ollama pull qwen2.5:7b-instruct`.
VS Build Tools	Only if better-sqlite3 ever has to compile from source; `go.ps1` says so.

> **Two install traps, already paid for.** npm 12 blocks install scripts, so better-sqlite3's native binary can fail silently while `npm install` reports success; `package.json` allowlists it in `allowScripts` and the launchers check it loads. On versions below v13, a newer Node had no prebuilt binary and the install fell back to a compiler.

08 Configuration and running

VARIABLE	DEFAULT	PURPOSE
`DB_PATH · AUDIO_DIR`	`./data/clinicos.db · ./data/audio`	Database file; audio until it is deleted at signing.
`ASR_PROVIDER · ASR_LANGUAGE`	`local` · empty	Local sidecar or an http endpoint; empty language means auto-detect.
`LLM_PROVIDER · LLM_OLLAMA_MODEL`	`ollama · qwen2.5:7b-instruct`	Local Ollama or an OpenAI-compatible endpoint; the drafting model.
`CLINIC_* · DOCTOR_*`	`Demo Clinic · Dr. Demo`	Fallback prescription identity until Clinic admin sets it.

Run: `./go.sh` on macOS, `powershell -ExecutionPolicy Bypass -File .\go.ps1` on Windows. Both are idempotent: they install what is missing, check the SQLite binary loads, create the Python virtualenv, find Ollama and start all three processes. **Test:** `npm run test:e2e` builds the app against a throwaway database (never `data/`) and runs 153 checks over HTTP.

09 Public website — a separate app

`landing/` is its own static Next.js app with no database, no API routes and no access to `clinicos-app`. That is why it can be public on Vercel while the clinical app never is. Runtime: next, react, react-dom, gsap, lenis, motion. Its backdrop film is prepared offline by `scripts/backdrop.py` with Real-ESRGAN, RIFE, imageio-ffmpeg, NumPy and Pillow. Those are build tools only: the browser receives AVIF frames drawn on a canvas.

10 Non-negotiable invariants

1. **No clinical judgement.** No diagnoses, risk scores or triage flags.
2. **Identifiable data never leaves the machine.**
3. **Audio is deleted when the doctor signs.**
4. **Never train on patient audio.**
5. **Consent is enforced server-side.** 403 without it.
6. `draftJson` **is immutable.** Edits go to `workingJson`.
7. **Backups use SQLite's online backup API.**
8. **No lock-in.** Full export in an open format.